

Nathcorp Internship Project Task Document

Project Title: AI Medical Chatbot

1. Project Task

You are required to build a basic AI Medical Information Chatbot that can answer general health-related questions.

The chatbot should provide simple information about common diseases, symptoms, medicines, precautions, emergency warning signs, and uploaded medical PDFs such as prescriptions or lab reports.

This project is only for learning and demonstration purposes. The chatbot must not provide medical diagnosis, prescription, dosage advice, or treatment decisions.

2. Project Goal

The goal of this project is to help interns understand how to build a chatbot-based application using frontend, backend, optional API integration, and RAG-based PDF question answering.

By completing this project, interns will learn:

- How to create a chatbot user interface.
- How to build backend APIs.
- How to process user questions.

- How to integrate free public medical APIs or trusted health information sources.
- How to upload and process PDF documents.
- How to use RAG to answer questions from uploaded prescription or report PDFs.
- How to apply safety rules in a healthcare-related chatbot.

3. Main Features to Build

3.1 Chatbot Interface

Interns need to create a simple chatbot screen where users can type health-related questions.

The chatbot should:

- Accept user questions.
- Display chatbot responses clearly.
- Show a medical safety disclaimer where required.
- Keep the UI simple and user-friendly.

3.2 General Health Information

The chatbot should answer basic health-related questions.

Example questions:

- What is diabetes?
- What are the symptoms of dengue?
- What is fever?
- How can I prevent malaria?
- What precautions should I take for cold and cough?
- What is high BP?

The response should be simple, clear, and easy to understand.

3.3 Disease Information

Interns need to add support for common disease-related queries.

The chatbot should provide:

- *Basic disease explanation.*
- *Common symptoms.*
- *General causes.*
- *Prevention tips.*
- *When to consult a doctor.*

Example diseases:

- *Diabetes*
- *Asthma*
- *Dengue*
- *Malaria*
- *Fever*
- *Migraine*
- *Cold and cough*
- *High BP / High Blood Pressure*

3.4 Medicine Information

The chatbot should provide only basic medicine-related information.

It can include:

- *General usage of medicine.*
- *Basic warnings.*
- *Common medicine information.*

The chatbot must not provide:

- Dosage advice.
- Prescription.
- Recommendation to start or stop medicine.
- Personal medical decision.

Example question:

- What is paracetamol used for?

3.5 Emergency Symptom Detection

Interns need to add logic to identify serious symptoms.

Emergency symptoms may include:

- Chest pain
- Breathing difficulty
- Heavy bleeding
- Unconsciousness
- Stroke symptoms
- Severe allergic reaction
- Very high blood pressure with severe headache, chest pain, blurred vision, or breathing difficulty.

For such cases, the chatbot should immediately show a warning message:

“This may be a serious condition. Please contact a doctor or emergency medical service immediately.”

3.6 PDF Upload and RAG-Based Question Answering

Interns need to add a feature where users can upload a medical PDF such as a prescription, lab report, or health report.

After uploading the PDF, the chatbot should allow users to ask questions related to that document.

Example PDF-related questions:

- *What medicines are mentioned in this prescription?*
- *What tests are mentioned in this report?*
- *Summarize this medical report in simple language.*
- *What are the important points from this prescription?*
- *Which values are marked high or low in this report?*
- *What should I ask my doctor based on this report?*

The chatbot should answer only based on the uploaded PDF content.

The chatbot should not:

- *Diagnose the user based on the PDF.*
- *Suggest treatment based on the report.*
- *Change or recommend medicine dosage.*
- *Say whether the report is normal or dangerous without advising doctor consultation.*
- *Replace a doctor's interpretation.*

The chatbot should include a disclaimer such as:

“This answer is generated from the uploaded document for informational purposes only. Please consult a licensed doctor for proper medical interpretation.”

4. Suggested Technology Stack

Interns can use the following technologies:

- *Frontend: React.js, HTML, CSS, Bootstrap, or Tailwind CSS.*
- *Backend: Python FastAPI.*
- *PDF Processing: PyMuPDF, pdfplumber, or pypdf.*
- *RAQ Framework: LangChain or LlamaIndex.*
- *Vector Store: FAISS or ChromaDB.*

- *Embeddings: Sentence Transformers or any free local embedding model.*
- *Optional APIs: Free public medical APIs or trusted public health information sources.*
- *Optional AI Model: Local LLM using Ollama or any free/open-source model.*
- *Optional Storage: Browser local storage or simple file-based storage for testing logs.*

5. Safety Rules

Interns must follow these safety rules while developing the chatbot:

- *The chatbot must not diagnose any disease.*
- *The chatbot must not prescribe medicines.*
- *The chatbot must not suggest medicine dosage.*
- *The chatbot must not ask users to start, stop, or change medicine.*
- *The chatbot must always suggest consulting a licensed doctor for medical advice.*
- *For emergency symptoms, the chatbot must advise users to contact emergency services immediately.*
- *For uploaded PDFs, the chatbot must answer only from the uploaded document content.*
- *The chatbot must not make assumptions about missing information in the PDF.*
- *The chatbot must not store unnecessary personal or medical information.*
- *Uploaded documents should be handled carefully because they may contain sensitive health information.*

6. Sample Test Prompts

Use the following prompts for testing:

- What is diabetes?
- What are the symptoms of dengue?
- What is high BP?
- What are the symptoms of high blood pressure?
- What precautions should I take for high BP?
- What is paracetamol used for?
- I have chest pain and breathing difficulty.
- How can I prevent malaria?
- What precautions should I take for cold and cough?
- Upload a prescription PDF and ask: What medicines are mentioned in this prescription?
- Upload a lab report PDF and ask: Which tests are included in this report?
- Upload a report PDF and ask: Summarize this report in simple language.
- Upload a prescription PDF and ask: What should I ask my doctor about this prescription?

7. Development Resources

This project can be completed free of cost for internship and demo purposes.

The project can remain free if interns use free development tools, free public APIs, open-source RAG tools, and local system setup instead of paid cloud services.

Development Tools

- Visual Studio Code for coding.
- Git and GitHub for version control.
- Python for backend development.
- FastAPI for backend API creation.
- React.js for frontend development.
- HTML, CSS, Bootstrap, or Tailwind CSS for UI design.

- Postman or Thunder Client for API testing.
- SQLite, local JSON files, or browser local storage for simple testing storage.

Medical Information APIs and Sources

- MedlinePlus can be used for patient-friendly health information and disease-related content.
- RxNorm API can be used for medicine name lookup and drug information mapping.
- openFDA API can be used for public FDA drug labels, warnings, recalls, and adverse event data.
- ClinicalTrials.gov API can be used for clinical trial search and research-related information.
- PubMed / NCBI E-utilities can be used for searching medical research articles and abstracts.
- WHO public health data can be used for global health statistics and public health indicators.

RAG and PDF Processing Resources

- PyMuPDF can be used to extract text from PDF files.
- pdfplumber can be used to read text and tables from PDFs.
- pypdf can be used for basic PDF text extraction.
- FAISS can be used as a local vector database.
- ChromaDB, Pinecone, Datastax can be used as a local vector store.
- Sentence Transformers can be used for free local embeddings.
- LangChain can be used to build the RAG pipeline.
- LlamaIndex can be used to index documents and build document question-answering workflows.

Free Hosting Options for Demo

- Run frontend and backend locally on intern laptops.
- Use GitHub repository for code sharing.
- Use local FastAPI server for backend testing.
- Use local React development server for frontend demo.
- Optional free hosting platforms can be used only if required for final demonstration.

For internship purposes, the recommended approach is to build and run the project locally using free tools, free public APIs, and open-source RAG components but deployment would be better.

8. Medical Disclaimer

This chatbot provides general health information only. It does not provide medical diagnosis, treatment, prescription, report interpretation, or emergency care. Answers from uploaded PDFs are generated only from the document content and are for informational purposes. Users should consult a licensed doctor or healthcare professional for medical advice, prescription clarification, or report interpretation. In case of emergency symptoms, users should contact emergency medical services immediately.